Human Evolution

Chapter 19/20

Syllabus Statement

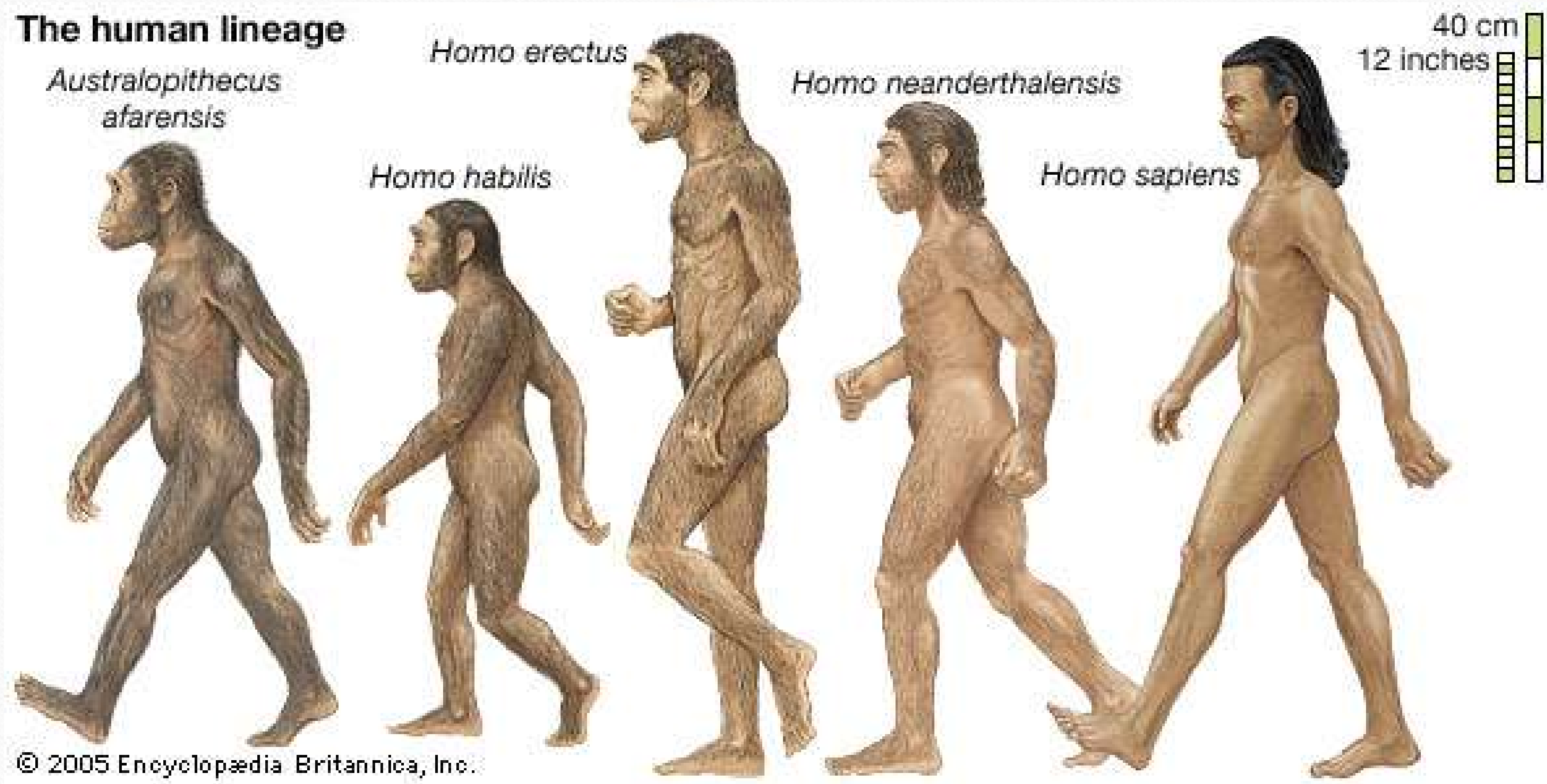
Create a timeline showing how human have evolved from other hominids

Describe evidence that suggests we are related to other hominids

Compare the features of;
- Australopithecus afarensis
- Australopithecus africanus
- Paranthropus robustus
- Homo habilis
- Homo erectus
- Homo neanderthalensis
- Homo sapiens

Summarize trends of tool use and how that links to cognitive/lifestyle development

Describe evidence of tool use in
- Homo habilis
- Homo erectus
- Homo neanderthalensis
- Homo sapiens



The human lineage
Australopithecus afarensis
Homo habilis
Homo erectus
Homo neanderthalensis
Homo sapiens
40 cm
12 inches
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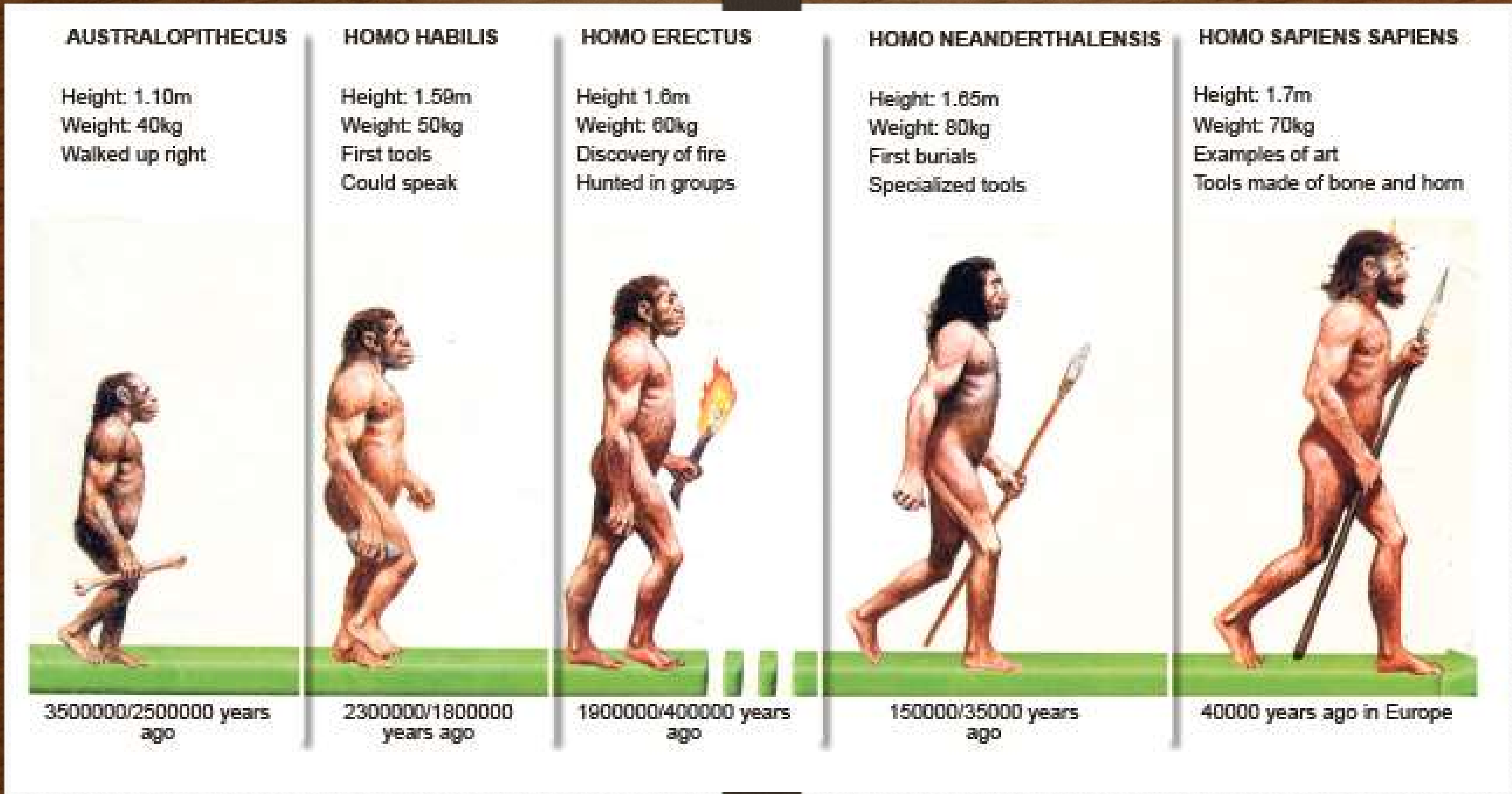
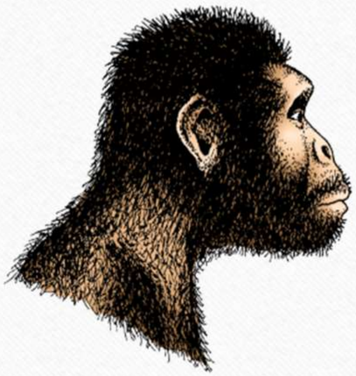
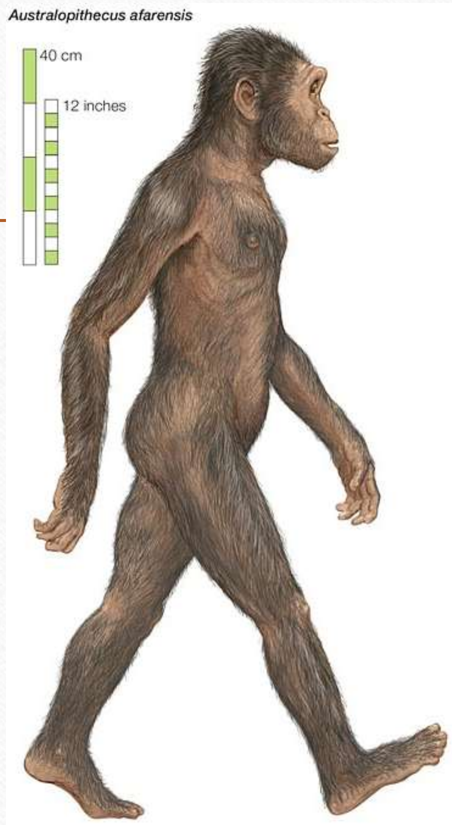
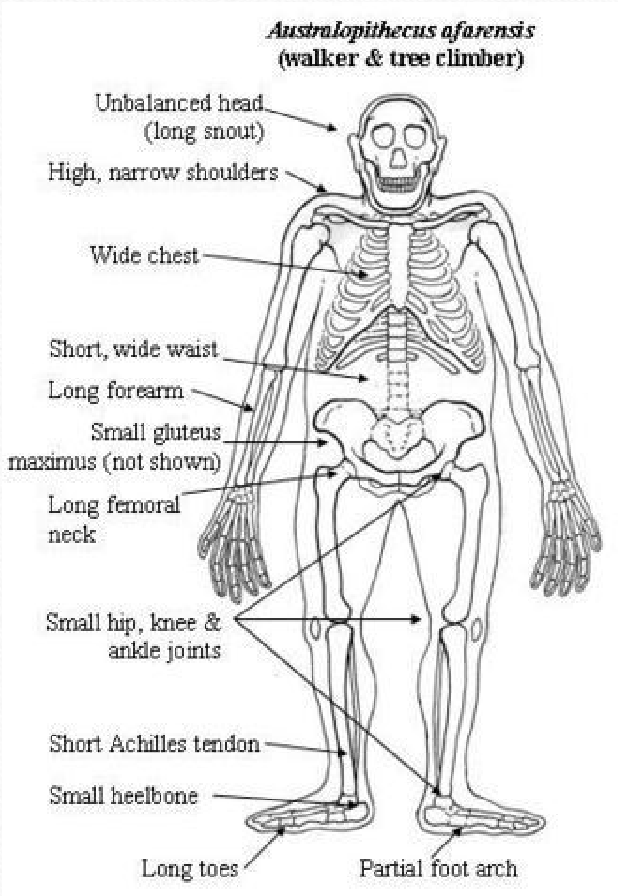

AUSTRALOPITHECUS
Height: 1.10m
Weight: 40kg
Walked up right
3500000/2500000 years ago
HOMO HABILIS
Height: 1.59m
Weight: 50kg
First tools
Could speak
2300000/1800000 years ago
HOMO ERECTUS
Height 1.6m
Weight: 60kg
Discovery of fire
Hunted in groups
1900000/400000 years ago
HOMO NEANDERTHALENSIS
Height: 1.65m
Weight: 80kg
First burials
Specialized tools
150000/35000 years ago
HOMO SAPIENS SAPIENS
Height: 1.7m
Weight: 70kg
Examples of art
Tools made of bone and horn
40000 years ago in Europe

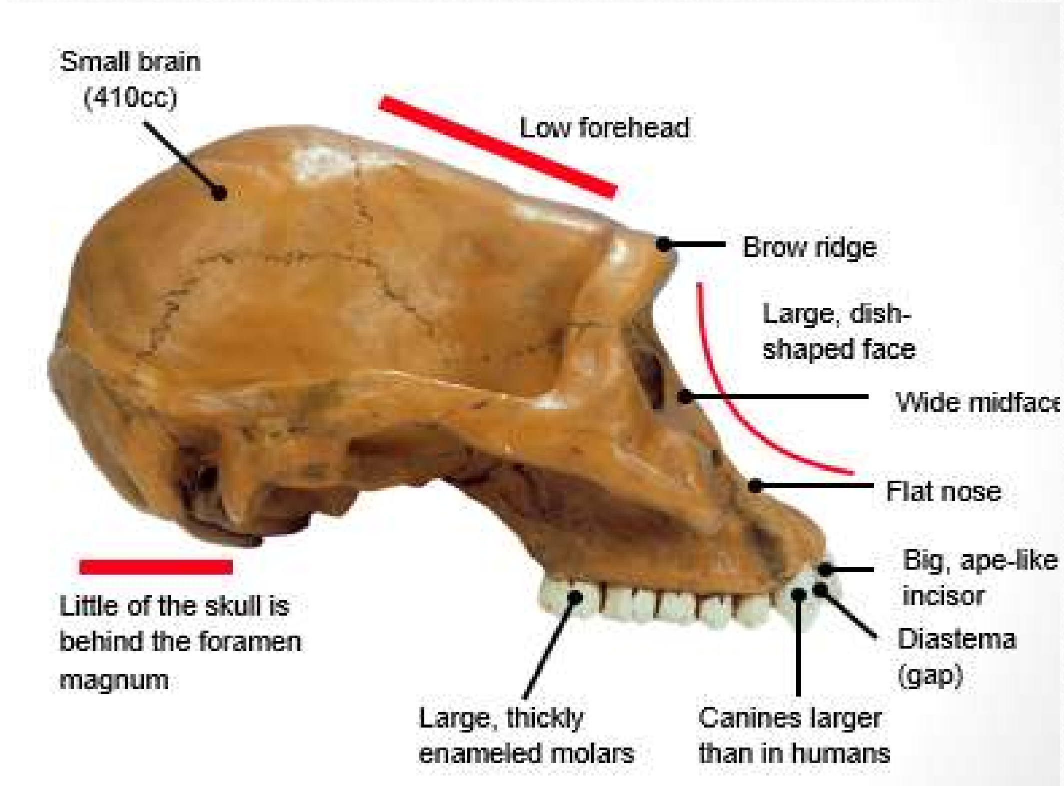
Australopithecus Afarensis

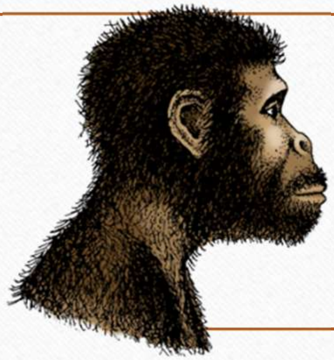
Nicknames	Lucy, first family
Age	3.9 - 3.0 mya
Location	Eastern Africa
Physique	- Light build - Long arms - Sexually dysmorphic
Height	1-1.5m





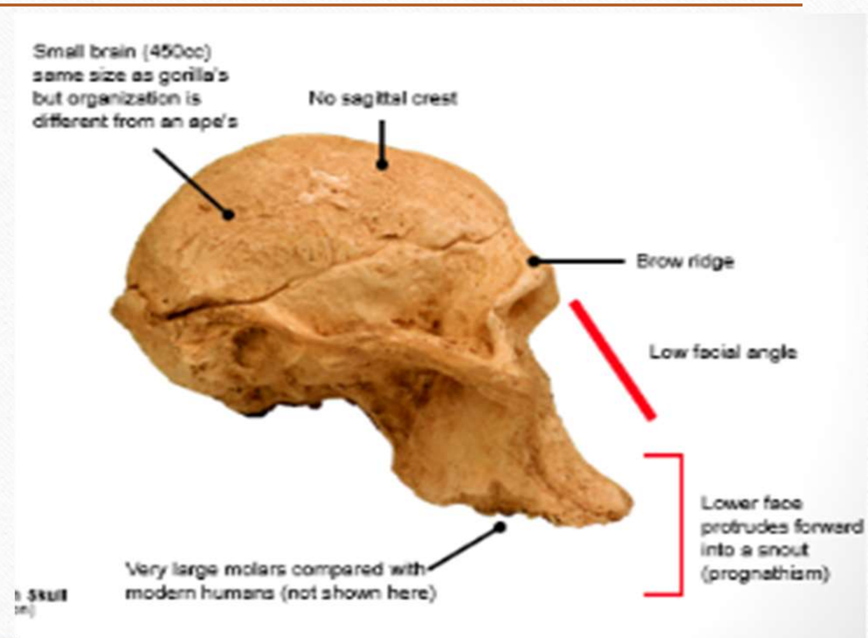

Australopithecus afarensis
(walker & tree climber)
Unbalanced head (long snout)
High, narrow shoulders
Wide chest
Short, wide waist
Long forearm
Small gluteus maximus (not shown)
Long femoral neck
Small hip, knee & ankle joints
Short Achilles tendon
Small heelbone
Long toes
Partial foot arch

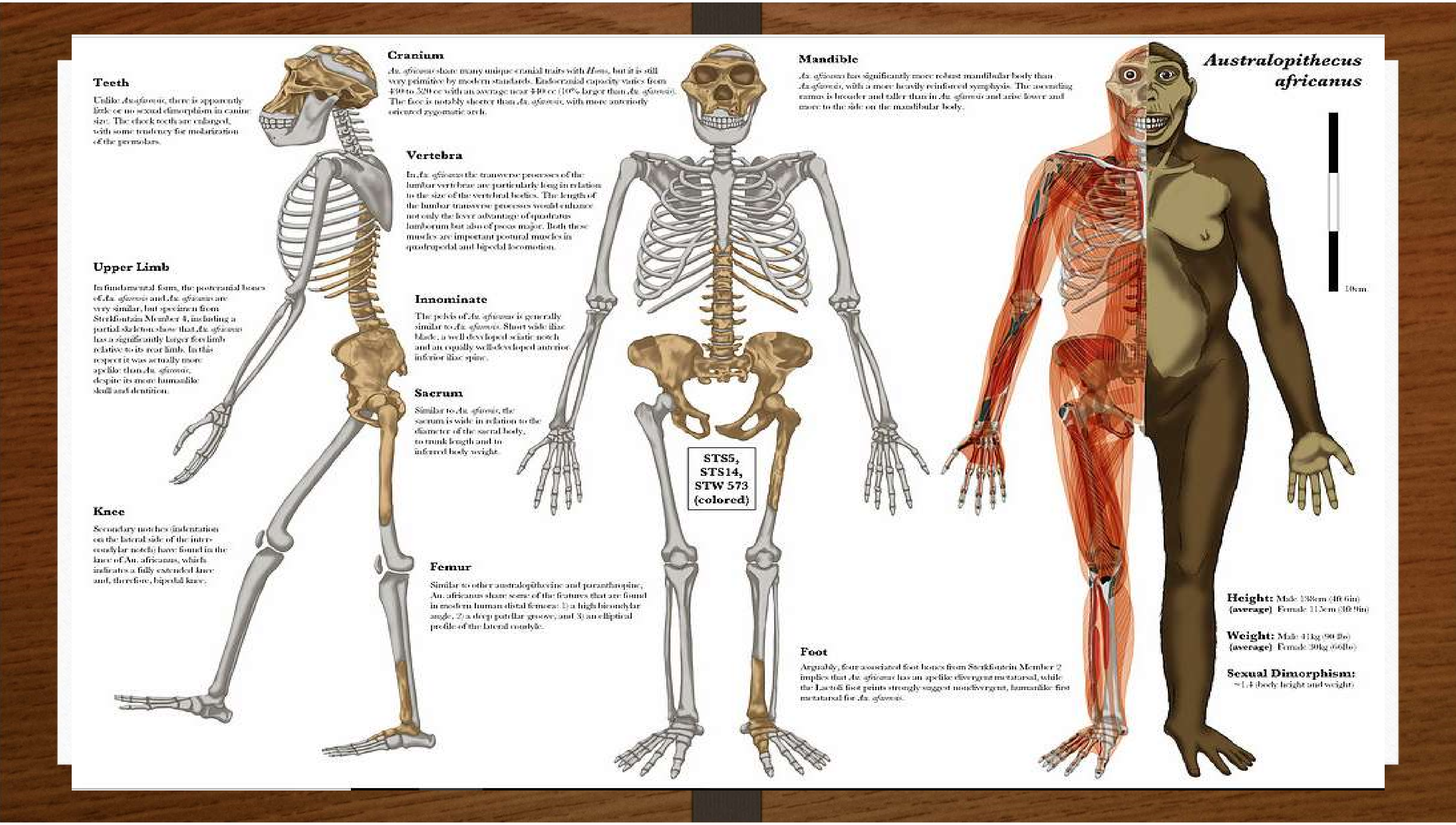

Small brain (410cc)
Low forehead
Brow ridge
Large, dish-shaped face
Wide midface
Flat nose
Big, ape-like incisor
Diastema (gap)
Little of the skull is behind the foramen magnum
Large, thickly enameled molars
Canines larger than in humans

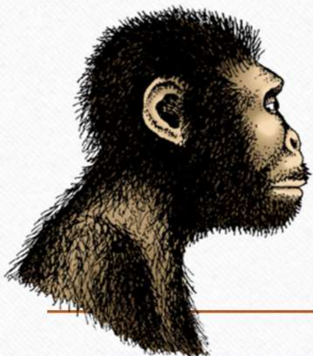


Australopithecus Africanis

Nicknames	
Age	3.9 - 3.0 myaq
Location	Southern Africa
Physique	More human like features than afarensis
Height	1.1-1.4m

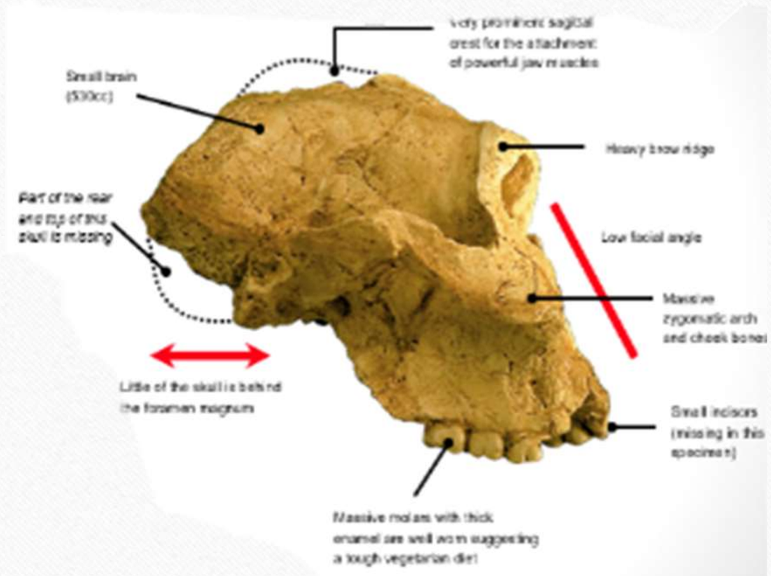



Teeth
Unlike Au. afarensis, there is apparently little or no sexual dimorphism in canine size. The cheek teeth are enlarged, with some tendency for molarization of the premolars.
Cranium
Au. africanus share many unique cranial traits with Homo, but it is still very primitive by modern standards. Endocranial capacity varies from 430 to 520 cc with an average near 440 cc (10% larger than Au. afarensis). The face is notably shorter than Au. afarensis, with more anteriorly oriented zygomatic arch.
Mandible
Au. africanus has significantly more robust mandibular body than Au. afarensis, with a more heavily reinforced symphysis. The ascending ramus is broader and taller than in Au. afarensis and arise lower and more to the side on the mandibular body.
Australopithecus africanus
Vertebra
In Au. africanus the transverse processes of the lumbar vertebrae are particularly long in relation to the size of the vertebral bodies. The length of the lumbar transverse processes would enhance not only the lever advantage of quadratus lumborum but also of psoas major. Both these muscles are important postural muscles in quadrupedal and bipedal locomotion.
Upper Limb
In fundamental form, the postcranial bones of Au. afarensis and Au. africanus are very similar, but specimen from Sterkfontain Member 4, including a partial skeleton show that Au. africanus has a significantly larger forelimb relative to its rear limb. In this respect it was actually more apelike than Au. afarensis, despite its more humanlike skull and dentition.
Innominate
The pelvis of Au. africanus is generally similar to Au. afarensis. Short wide iliac blade, a well developed sciatic notch and an equally well-developed anterior inferior iliac spine.
Sacrum
Similar to Au. afarensis, the sacrum is wide in relation to the diameter of the sacral body, to trunk length and to inferred body weight.
STS5, STS14, STW 573 (colored)
10cm
Knee
Secondary notches (indentation on the lateral side of the inter-condylar notch) have found in the knee of Au. africanus, which indicates a fully extended knee and, therefore, bipedal knee.
Femur
Similar to other australopithecine and paranthropine, Au. africanus share some of the features that are found in modern human distal femora: 1) a high bicondylar angle, 2) a deep patellar groove, and 3) an elliptical profile of the lateral condyle.
Height: Male 138cm (4ft 6in)
(average) Female 115cm (3ft 9in)
Weight: Male 41kg (90 lbs)
(average) Female 30kg (66lbs)
Sexual Dimorphism:
~1.4 (body height and weight)
Foot
Arguably, four associated foot bones from Sterkfontein Member 2 implies that Au. africanus has an apelike divergent metatarsal, while the Laetoli foot prints strongly suggest nondivergent, humanlike first metatarsal for Au. afarensis.

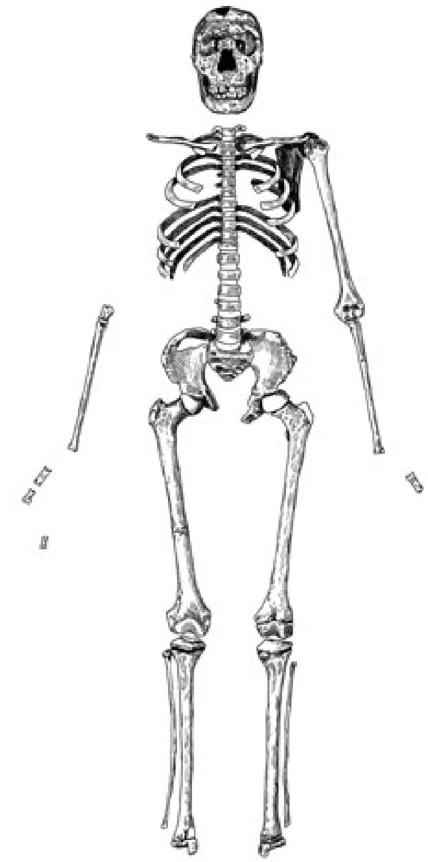


Paranthropus Robustus

Nicknames	Megadon
Age	2.0 - 1.5 mya
Location	Southern Africa
Physique	More robust and heavy set
Height	1.1-1.3m

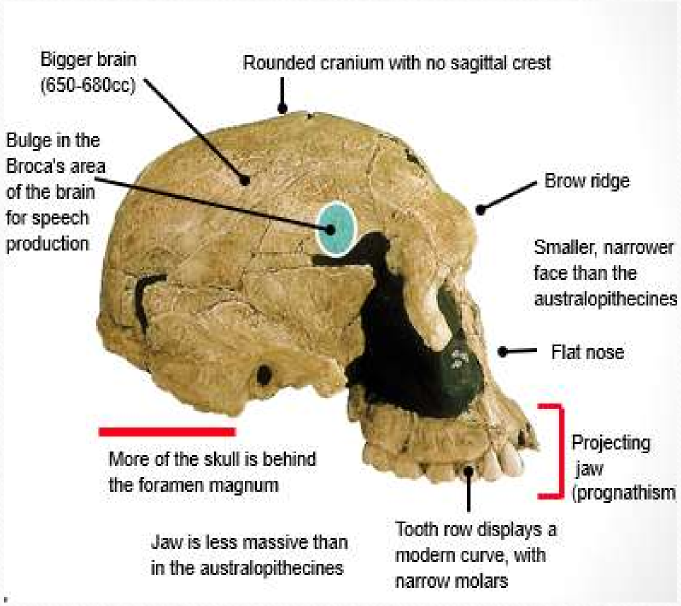


AUSTRALOPITHECUS ROBUSTUS FEMALE



Homo Habilis

Nicknames	
Age	2.4 – 1.5 mya
Location	Eastern Africa
Physique	Long arms
Height	1.0m



Homo Habilis appeared 2.4 million years ago

Homo Habilis was the first group to make stone tools. They were called the Handy Man because of it

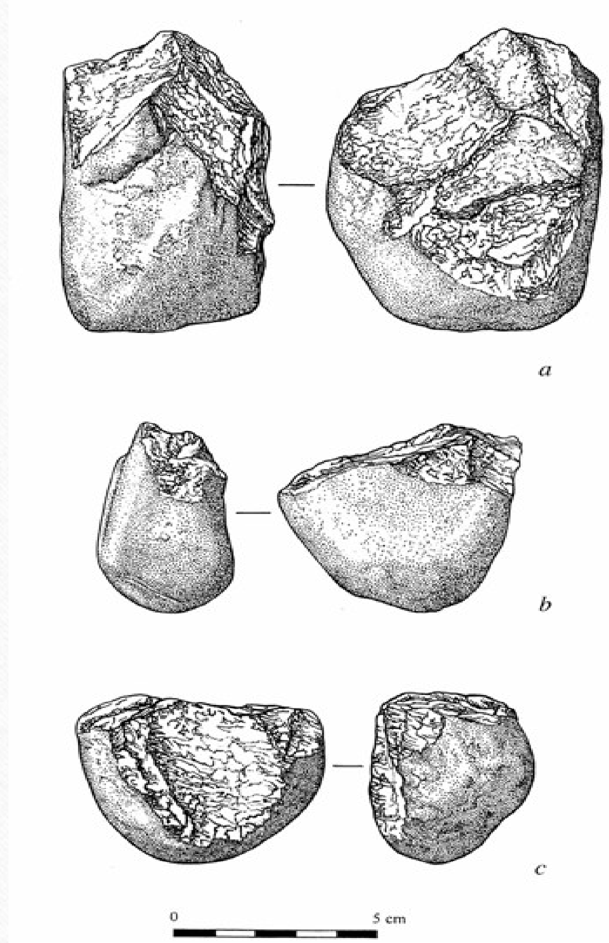
Homo Habilis disappeared 1.4 million years ago

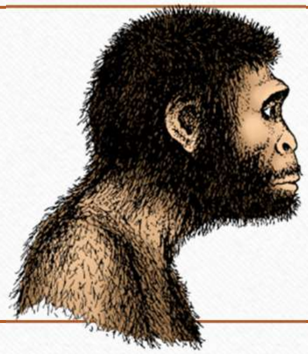
Appearance

While They Lived

Disappearance

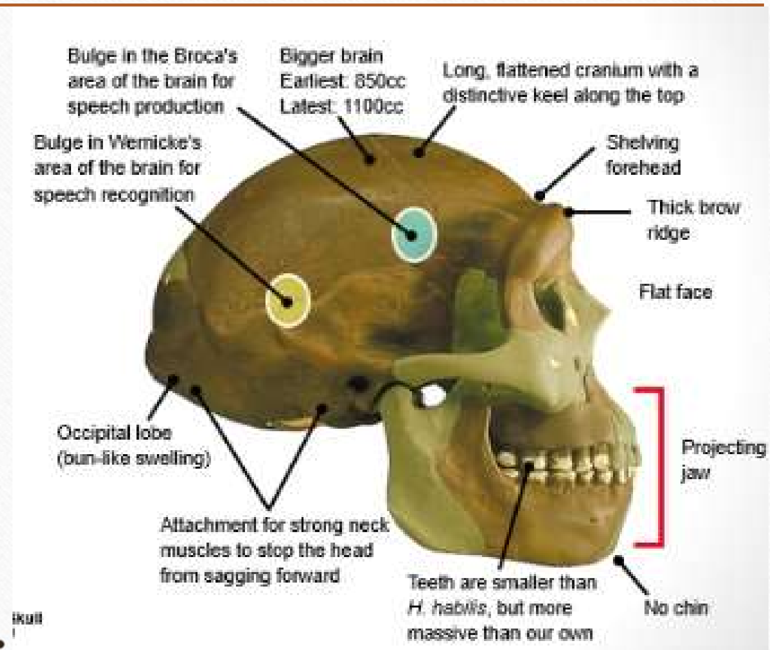
Cultural Period	**Oldowan**
Tool Material	River worn pebbles that were crudely shaped. Few flakes taken off.
Making Method	Hard hammer percussion
Types	Choppers, bi face.
Communication	Facial expression and guestures for language
Shelter	12 people, made shelters
Hunting	Meat eating diet. Food sharing

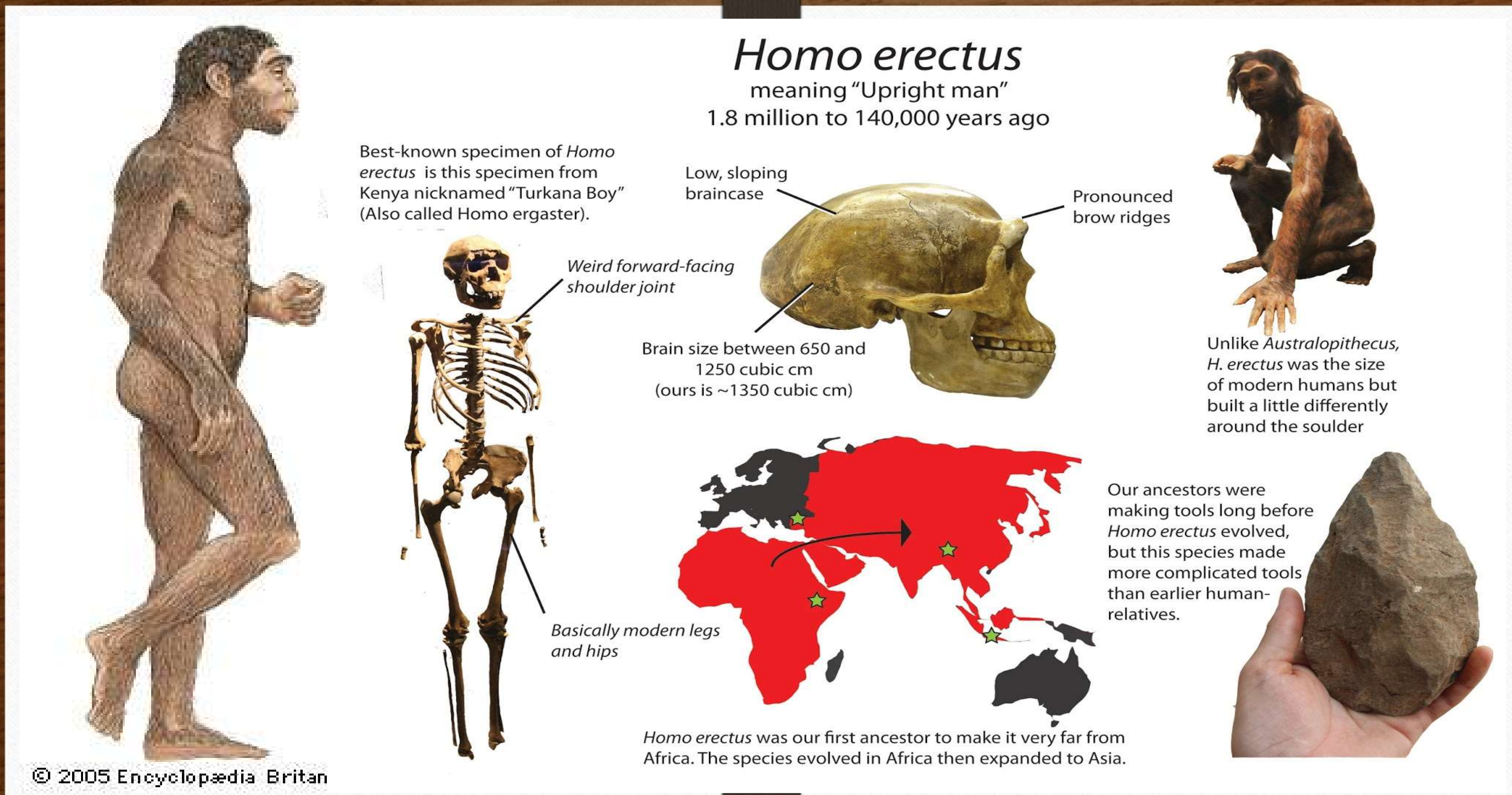




Homo Erectus

Nicknames	
Age	1.8 – 0.3 mya
Location	Africa, Asia, Indonesia, Europe
Physique	Robust but human like skeleton
Height	1.3-1.5m




Homo erectus
meaning "Upright man"
1.8 million to 140,000 years ago
Best-known specimen of Homo erectus is this specimen from Kenya nicknamed "Turkana Boy" (Also called Homo ergaster).
Weird forward-facing shoulder joint
Basically modern legs and hips
Low, sloping braincase
Pronounced brow ridges
Brain size between 650 and 1250 cubic cm (ours is ~1350 cubic cm)
Unlike Australopithecus, H. erectus was the size of modern humans but built a little differently around the soulder
Our ancestors were making tools long before Homo erectus evolved, but this species made more complicated tools than earlier human-relatives.
Homo erectus was our first ancestor to make it very far from Africa. The species evolved in Africa then expanded to Asia.
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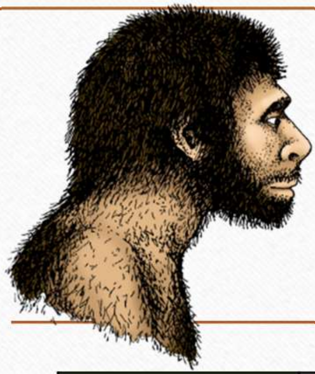
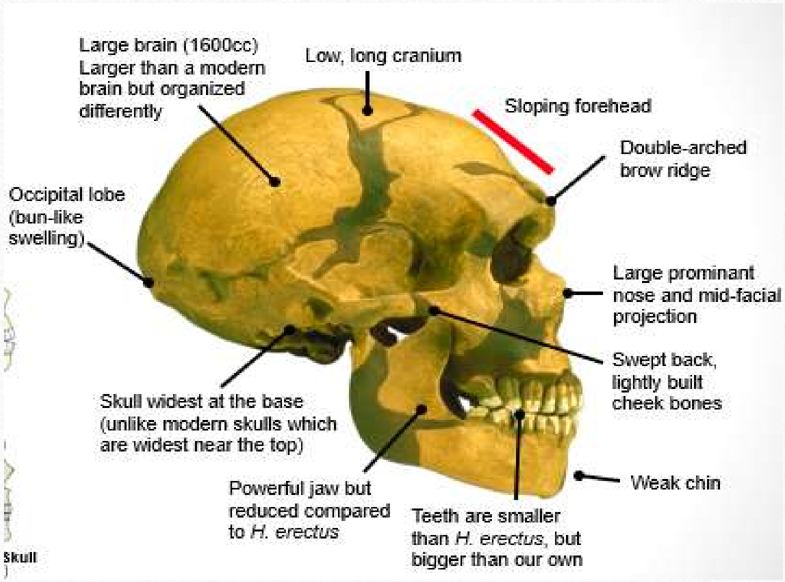
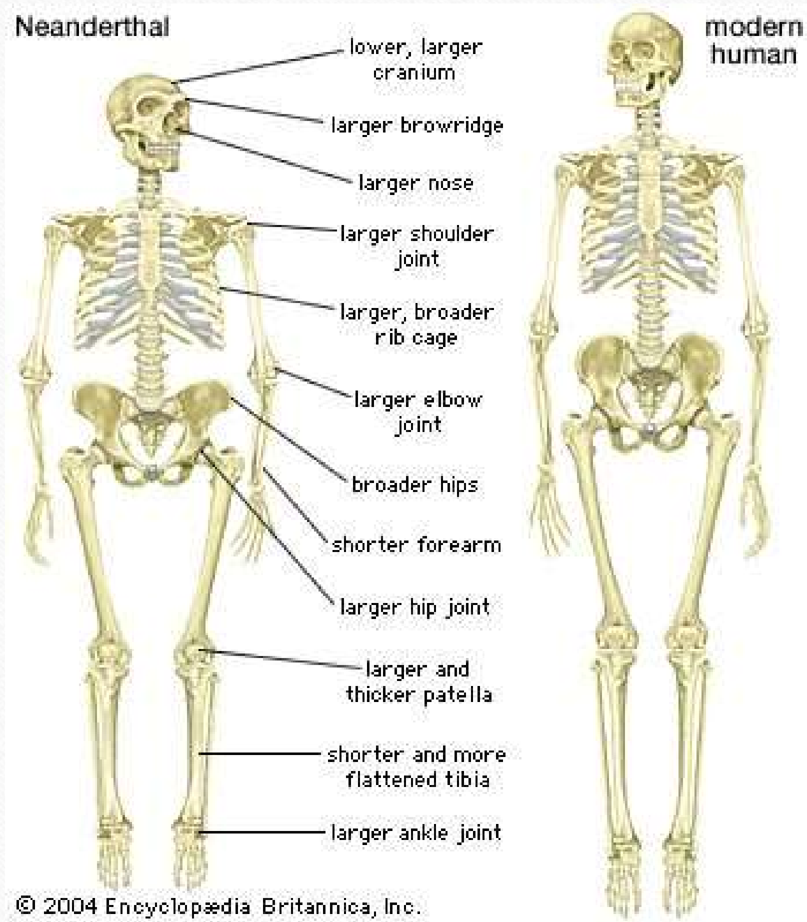
Cultural Period	**Acheulian**
Tool Material	Mainly pebbles and stone
Making Method	Levallous making (refined hard hammer percussion)
Types	Bi faced, hand ax
Fire	First traces of controlled fire
Shelter	Huts wooden. First evidence of camps
Hunting	Cooperative. Gathering was 70% of diet with the ability to catch large prey



Homo Neanderthal

Nicknames	Cold Cave Man
Age	0.23-0.028 mya
Location	Europe, west Africa
Physique	Robust but more adaptations for the cold
Height	1.5-1.7m




Neanderthal
lower, larger cranium
larger browridge
larger nose
larger shoulder joint
larger, broader rib cage
larger elbow joint
broader hips
shorter forearm
larger hip joint
larger and thicker patella
shorter and more flattened tibia
larger ankle joint
modern human
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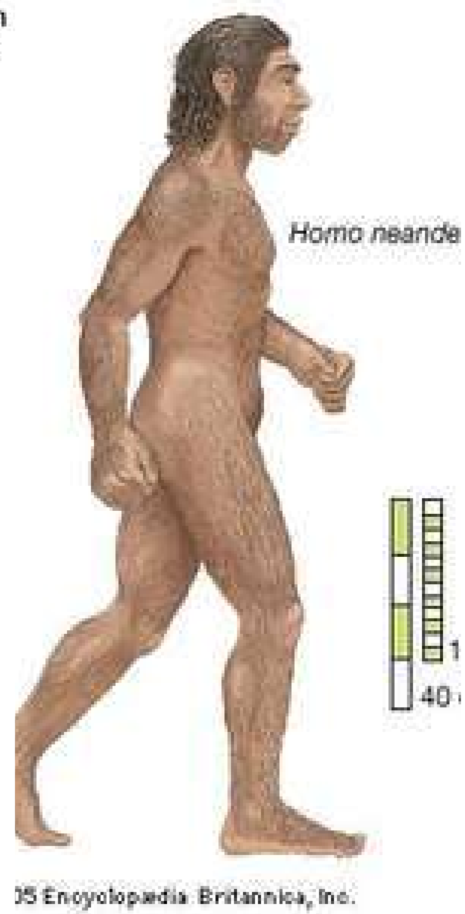
Homo neande
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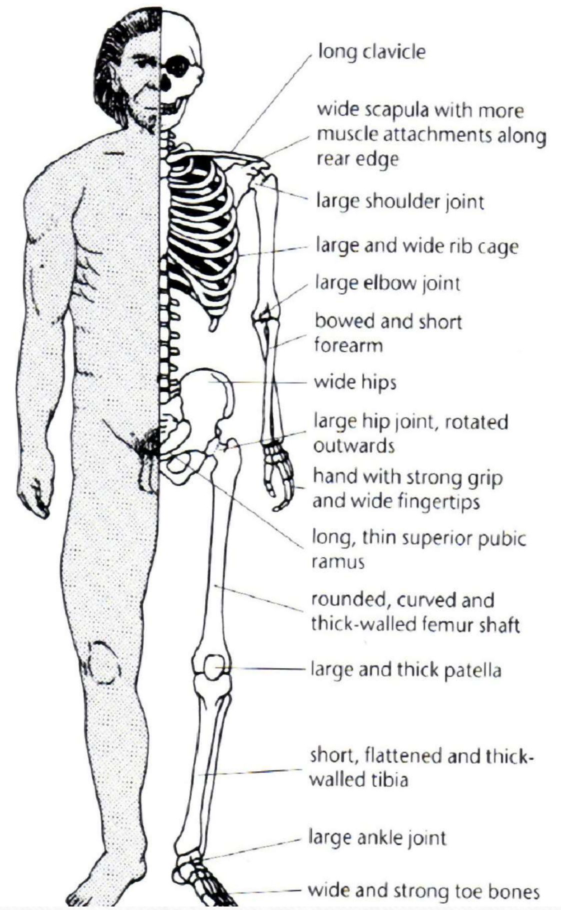
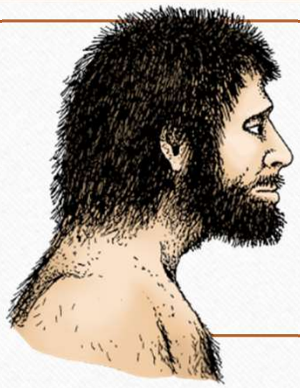
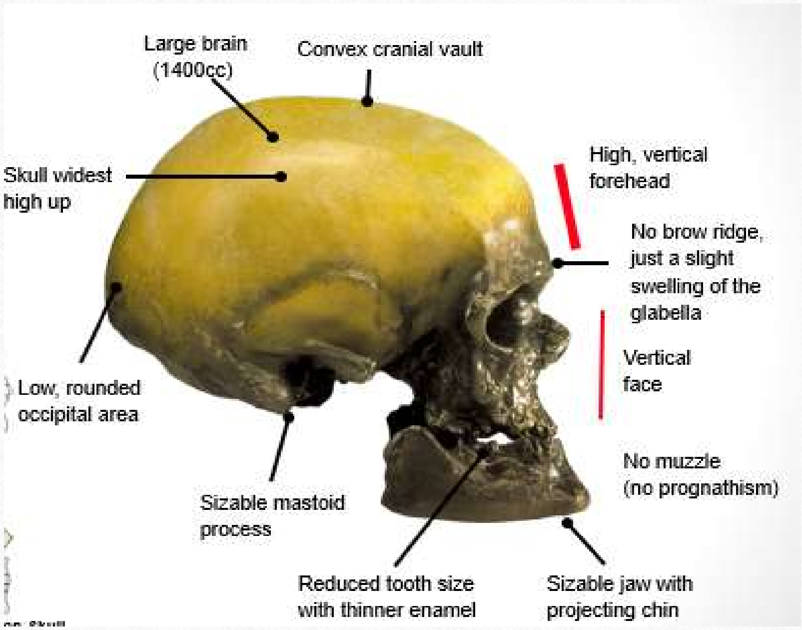

long clavicle
wide scapula with more muscle attachments along rear edge
large shoulder joint
large and wide rib cage
large elbow joint
bowed and short forearm
wide hips
large hip joint, rotated outwards
hand with strong grip and wide fingertips
long, thin superior pubic ramus
rounded, curved and thick-walled femur shaft
large and thick patella
short, flattened and thick-walled tibia
large ankle joint
wide and strong toe bones

Cultural Period	**Mousterian**
Tool Material	Flint was seeked out as its easy to work with
Making Method	Finer workman ship included soft hammer percussion using bone. Levallous making
Types	Axes and scrapers
Communication	Strong social bondings
Art	Music – flute found. Bone jewellery
Shelter	Caves with stone walls



Homo Sapiens

Nicknames	Modern day humans
Age	160000 ya
Location	Earliest in Asia and Africa
Physique	Modern skeleton
Height	1.6-1.85m



Cultural Period	**Upper Paleolithic** (**A**uriacian, **M**agdalenian, **S**olutrean)
Tool Material	Antlers, bone, stick
Making Method	Punch blade method. Carving.
Types	Throwing sticks, needles,
Communication	Large groups and language developed
Art	Sewing, paintings, engraving, singing
Hunting	Highly skilled hunters